

Hankel-Structured Gerchberg–Saxton Channel Estimation for Rydberg Atomic MIMO Receivers

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Abstract—Rydberg atomic quantum receivers measure only the *magnitude* of the incident field superposed on a local reference, making multi-user channel estimation a biased phase-retrieval rather than a linear-Gaussian problem. The known solvers—biased Gerchberg–Saxton (GS) and its expectation–maximisation refinement EM-GS—treat the channel as unstructured, and their normalised mean-square error (NMSE) consequently does not improve as the atomic array grows. We observe that on a uniform linear array each propagation path contributes exactly one spatial complex exponential, so the Hankel lifting of every channel column has rank at most the path count. Interleaving a Cadzow low-rank Hankel projection with every exact EM-GS update yields HS-GS; the model order comes from a held-out pilot residual, no oracle path count is used, and the magnitude observation is never linearised. We also derive unconstrained and geometry-constrained Cramér–Rao bounds for the exact Rician likelihood. Over 3.6×10^4 paired trials, and under the tested sparse geometric multipath model with i.i.d. uniform angles of arrival and equal per-path power, HS-GS improves on EM-GS by a mean of 2.85 dB at $N = 32$. Plotting the two error curves separately shows why: EM-GS is flat in array size to within 0.04 dB, so HS-GS is the only estimator whose NMSE improves as the aperture grows, and the gap widens monotonically ($0.03 \rightarrow 0.81 \rightarrow 2.45$ dB at $N = 8/16/32$). The benefit is conditional rather than general: at $N = 8$ the projection helps by 0.74 dB at -10 dB SNR but *hurts* by up to 0.40 dB above 0 dB. A controlled path-count experiment confirms the mechanism: the gain decays monotonically from 7.04 dB at $L = 2$ to zero exactly as the path count reaches the Hankel rank ceiling. Robustness under clustered propagation is not established and is the decisive next test.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, Cadzow projection, constrained Cramér–Rao bound.

I. INTRODUCTION

Rydberg atomic quantum receivers replace the RF front end with optically read excited atoms, giving sensitivity and bandwidth flexibility unavailable to conventional antennas [1], [2]. Cui *et al.* [1] showed the resulting atomic MIMO receiver does not obey the linear transmission model: heterodyne readout against a local reference yields the *modulus* of the summed field, making recovery a *biased phase-retrieval* problem [3], [4]. They adapted Gerchberg–Saxton (GS) [5] to this setting and proposed EM-GS, replacing the hard modulus substitution with the exact posterior mean of the phase factor.

Both solvers treat $\mathbf{G} \in \mathbb{C}^{N \times K}$ as $2NK$ free real parameters. Under the magnitude model the Fisher information is block diagonal across receive elements (Section V), so estimation separates into N per-row problems: enlarging the array adds

as many unknowns as measurements and leaves the *normalised* error unchanged—EM-GS varies by only 0.012 dB across $N \in \{8, 16, 32\}$. Any improvement scaling with array size must come from structure coupling the rows.

Such structure exists whenever the array is a ULA and propagation is via a few resolvable paths: each path imprints one spatial complex exponential across the aperture. Grid-based and subspace estimators exploit this in conventional MIMO, but it has not been used inside a magnitude-only atomic receiver, where the non-linearity forbids linear least squares.

Contributions. We show the Hankel lifting of each ULA channel column has rank at most L_k , turning path sparsity into a grid-free low-rank constraint needing no angle estimation (§III); propose HS-GS, interleaving a Cadzow projection with every exact EM-GS update at an order chosen from a held-out pilot residual, not oracle knowledge (§IV); derive the unconstrained and geometry-constrained CRLBs for this Rician likelihood, correcting a rank-2 misconception of the per-measurement Fisher information (§V); and isolate *when* the method works with a controlled path-count experiment (§VII–VIII).

Notation. $(\cdot)^T$ and $(\cdot)^H$ are transpose and conjugate transpose; $|\cdot|$ on a matrix is elementwise modulus; $\odot$ is the Hadamard product; $\|\cdot\|_F$ is the Frobenius norm; I_ν is the order- ν modified Bessel function of the first kind.

II. SYSTEM MODEL

A. Magnitude-Only Measurement

K single-antenna users transmit to an N -element atomic ULA over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$ the channel, $\mathbf{S} \in \mathbb{C}^{K \times P}$ the pilots, $\mathbf{B} \in \mathbb{C}^{N \times P}$ the reference and $\mathbf{W} \in \mathbb{C}^{N \times P}$ the noise, the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}| \in \mathbb{R}_{\geq 0}^{N \times P}, \quad (1)$$

with $\mathbf{W}$ i.i.d. $\mathcal{CN}(0, \sigma^2)$ added *inside* the modulus. The phase of $\mathbf{GS} + \mathbf{B}$ is unobservable; only $\mathbf{B}$ makes $\mathbf{G}$ identifiable, through the cross term of $|x + b|^2 = |x|^2 + 2\Re\{xb^*\} + |b|^2$. Following [1], $\mathbf{B}$ is constant over (n, p) with zero phase, the SNR fixes

$$\sigma^2 = K/\text{SNR}_{\text{lin}}, \quad (2)$$

and the reference-to-signal ratio (RSR) is defined against a *single* user, giving $|b| = \sqrt{\text{RSR}_{\text{lin}}}$. Since K users transmit at

TABLE I
SIMULATION PARAMETERS. ALL EXPERIMENTS USE THE EXACT
OBSERVATION (1); THE MODULUS IS NEVER LINEARISED.

Channel and array		Estimators	
N	8, 16, 32	t_0	50 (GS, EM-GS)
K	3	T	50 (HS-GS)
L_k	$\mathcal{U}\{3, \dots, 7\}$ i.i.d.	project every	1 iteration
$\theta_{\ell,k}$	$\mathcal{U}[-\pi/2, \pi/2]$	n_{cz}	4 sweeps
$\alpha_{\ell,k}$	$\mathcal{CN}(0, \beta_k/L_k)$, $\beta_k=1$	ν	0.3 held out
c	1	T_{sel}	20
spacing	$\lambda/2$	$\hat{L}$ grid	$1, \dots, r_{\text{max}}$
Pilots and operating point		Monte Carlo	
P	10, 30 (swept 6–40)	trials/pt	200–1200, tiered
$\mathbf{S}$	$\mathcal{CN}(0, 1)$, redrawn	total	3.6×10^4
	full row rank, $P \geq 2K$	bootstrap	2000, paired
$\mathbf{B}$	constant, $\angle b = 0$		
SNR	–5 to 20 dB, (2)		
RSR	12 dB (swept 0–24)		

once, a nominal 12 dB RSR is 7.15 dB against the K -user total; all RSR values here are the single-user ones.

B. Geometric ULA Channel

The channel is generated as sparse geometric multipath. With half-wavelength spacing the electrical angle of path ℓ of user k is $\psi_{\ell,k} = \pi \sin \theta_{\ell,k}$, and

$$g_{n,k} = c \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad n = 1, \dots, N, \quad (3)$$

with atomic gain $c = 1$ (so the electrical and atomic channels coincide), $\alpha_{\ell,k} \sim \mathcal{CN}(0, \beta_k/L_k)$ and $\beta_k = 1$.

The assumptions used throughout: sparse geometric multipath with $L_k \sim \mathcal{U}\{3, \dots, 7\}$ i.i.d. per user per realisation; angles of arrival i.i.d. *uniform in θ* (Kolmogorov–Smirnov over 4465 samples: $p = 0.150$ against uniform-in- θ , $p < 10^{-4}$ against uniform-in-sin θ); equal per-path power; pilots i.i.d. $\mathcal{CN}(0, 1)$, redrawn each trial and rejection-sampled to full row rank with $P \geq 2K$, hence *not* orthogonal (largest off-diagonal $|\mathbf{S}\mathbf{S}^H|/P$ is 0.233); and flat, narrowband propagation. The channel is *not* a clustered 3GPP TR 38.901 or Saleh–Valenzuela model; Section VIII addresses the consequences. Table I collects the parameters.

C. Baseline Solvers

Row n of (1) depends only on row n of $\mathbf{G}$, so it reduces to N independent biased PR problems of dimension K . Biased GS [1] alternates a least-squares update with the hard modulus substitution $\hat{\mathbf{y}} = \mathbf{Z} \odot e^{j\angle(\mathbf{G}\mathbf{S}+\mathbf{B})}$; EM-GS instead uses the exact conditional mean of the phase factor, introducing the Bessel ratio

$$R(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \quad \kappa = \frac{2}{\sigma^2} \mathbf{Z} \odot |\mathbf{G}\mathbf{S} + \mathbf{B}|, \quad (4)$$

so the restored observation is shrunk by $R(\kappa) \in [0, 1]$ according to measurement reliability. As $\kappa \rightarrow \infty$, $R \rightarrow 1$ and EM-GS provably reduces to biased GS—a limit reached in practice at high RSR (§VII-A).

III. HANKEL STRUCTURE OF THE ULA CHANNEL

Drop the user index and write one channel column as $g_n = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^n$ with $z_{\ell} = e^{-j\psi_{\ell}}$. Define the Hankel lifting with pencil parameter p ,

$$\mathcal{H}(\mathbf{g}) = \begin{bmatrix} g_0 & g_1 & \cdots & g_p \\ g_1 & g_2 & \cdots & g_{p+1} \\ \vdots & & \ddots & \vdots \\ g_{N-p-1} & \cdots & & g_{N-1} \end{bmatrix} \in \mathbb{C}^{(N-p) \times (p+1)}. \quad (5)$$

Its (q, r) entry is g_{q+r} :

$$[\mathcal{H}(\mathbf{g})]_{q,r} = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^{q+r} = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^q z_{\ell}^r, \quad (6)$$

separating in q and r . With $\mathbf{v}_{\ell} = [1, z_{\ell}, \dots, z_{\ell}^{N-p-1}]^T$ and $\mathbf{w}_{\ell} = [1, z_{\ell}, \dots, z_{\ell}^{p-1}]^T$ this is

$$\mathcal{H}(\mathbf{g}) = \sum_{\ell=1}^L \alpha_{\ell} \mathbf{v}_{\ell} \mathbf{w}_{\ell}^T, \quad (7)$$

a sum of L rank-one terms, hence

$$\text{rank}(\mathcal{H}(\mathbf{g})) \leq L. \quad (8)$$

Each propagation path creates one spatial complex exponential across the ULA and therefore contributes exactly one rank-one component to the Hankel lifting, converting path sparsity into low-rank matrix structure without discretising the angular domain. When the z_{ℓ} are distinct and L does not exceed the matrix dimensions, (8) holds with equality, and conversely a sequence whose Hankel matrix has rank L is a sum of L exponentials [6].

The constraint has a ceiling: $\mathcal{H}(\mathbf{g})$ being $(N-p) \times (p+1)$, its rank cannot exceed $\min(N-p, p+1)$, maximised over p at

$$r_{\text{max}} = \max_{1 \leq p \leq N-1} \min(N-p, p+1) = \lceil N/2 \rceil, \quad (9)$$

so $r_{\text{max}} = 4, 8, 16$ for $N = 8, 16, 32$. If $L \geq r_{\text{max}}$ then rank $\leq L$ holds for every vector and carries no information—the ceiling states precisely *when* the prior is informative, and §VII-C tests it directly.

IV. PROPOSED HS-GS ESTIMATOR

A. Cadzow Projection

Let $\mathcal{S}_{\hat{L}}$ be the length- N sequences whose Hankel lifting has rank at most $\hat{L}$. Membership is the intersection of two sets with exact projectors: fixed-rank matrices (truncated SVD, Frobenius-optimal by Eckart–Young) and Hankel matrices (anti-diagonal averaging). Cadzow’s algorithm [6] alternates them: build $\mathbf{H} \leftarrow \mathcal{H}(\mathbf{g})$; take its SVD $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$; retain the $\hat{L}$ dominant components as $\mathbf{H}_{\text{lr}} = \mathbf{U}_{:,1:\hat{L}} \mathbf{\Sigma}_{1:\hat{L}} \mathbf{V}_{:,1:\hat{L}}^H$; restore Hankel structure by averaging each anti-diagonal, $\mathbf{g} \leftarrow \mathcal{H}^{\dagger}(\mathbf{H}_{\text{lr}})$; and repeat for $n_{cz} = 4$ sweeps. The intersection is a non-convex variety and four sweeps do not reach its exact nearest point, so $\mathcal{P}_{\hat{L}}$ is an *approximate* projection (Section VII-D quantifies the residual) and the iteration below carries no convergence guarantee. When $\hat{L} \geq r_{\text{max}}$ the routine returns its input unchanged, so HS-GS reproduces EM-GS bit for bit.

Algorithm 1 EM-GS measurement update $\mathcal{T}_{\text{EM}}$ (one iteration)

Require: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; iterate $\mathbf{G}$ ($\emptyset \Rightarrow$ spectral init)
1: $\mathbf{\Lambda} \leftarrow \mathbf{GS} + \mathbf{B}$; $\boldsymbol{\kappa} \leftarrow (2/\sigma^2) \mathbf{Z} \odot |\mathbf{\Lambda}|$
2: $\mathbf{R} \leftarrow R(\boldsymbol{\kappa})$, $R(x) = I_1(x)/I_0(x)$ $\triangleright$ elementwise
3: $\hat{\mathbf{Y}} \leftarrow \mathbf{R} \odot \mathbf{Z} \odot e^{j\angle \mathbf{\Lambda}}$ $\triangleright$ Rician posterior mean
4: $\mathbf{G}_{n,:} \leftarrow (\hat{\mathbf{Y}}_{n,:} - \mathbf{B}_{n,:})\mathbf{S}^H(\mathbf{SS}^H)^{-1}$, $n = 1..N$ $\triangleright$ rows independent
5: **return** $\mathbf{G}$

Algorithm 2 Cadzow projection $\mathcal{P}_{\hat{L}}$ of one channel column

Require: $\mathbf{g} \in \mathbb{C}^N$, rank $\hat{L}$, sweeps n_{cz}
1: $p \leftarrow \arg \max_p \min(N - p, p + 1)$; $r_{\text{max}} \leftarrow \lceil N/2 \rceil$
2: **if** $\hat{L} \geq r_{\text{max}}$ **then return** $\mathbf{g}$ $\triangleright$ constraint vacuous: exact no-op
3: **end if**
4: **for** $t = 1..n_{\text{cz}}$ **do**
5: $\mathbf{H} \leftarrow \mathcal{H}(\mathbf{g})$, $\mathbf{H}_{q,r} = g_{q+r}$ $\triangleright (N-p) \times (p+1)$
6: $\mathbf{U}\Sigma\mathbf{V}^H \leftarrow \text{svd}(\mathbf{H})$; $\mathbf{H} \leftarrow \mathbf{U}_{:,1:\hat{L}}\Sigma_{1:\hat{L}}\mathbf{V}_{:,1:\hat{L}}^H$ $\triangleright$ rank
7: $g_m \leftarrow |\mathcal{A}_m|^{-1} \sum_{(q,r) \in \mathcal{A}_m} \mathbf{H}_{q,r}$, $\mathcal{A}_m = \{q+r=m\}$ $\triangleright$ Hankel
8: **end for**
9: **return** $\mathbf{g}$

B. Interleaved Iteration and Order Selection

The projection acts *inside* the exact iteration, not as post-processing:

$$\mathbf{G}^{(t)} = \mathcal{P}_{\hat{L}} \circ \mathcal{T}_{\text{EM-GS}}(\mathbf{G}^{(t-1)}), \quad t = 1, \dots, T, \quad (10)$$

with $\mathcal{P}_{\hat{L}}$ acting column-wise after *every* one of the $T = 50$ updates. This matters because EM-GS update quality depends on the current iterate through κ in (4): a structurally cleaner iterate yields a better phase estimate, hence a better next iterate. Algorithms 1–3 give the measurement update, the projection, the rank selector and their composition; the paragraph after Alg. 3 states precisely which lines separate HS-GS from the baseline.

HS-GS needs a target rank $\hat{L}$, and the true L_k is unknown to any practical receiver. The in-sample residual cannot supply it: constraining $\mathbf{G}$ to a smaller set can only worsen the fit to the pilots used for fitting, so that criterion always prefers the largest candidate. Splitting the pilots into a fitting set and a validation set of fraction $\nu = 0.3$ penalises over-modelling correctly, and every quantity involved is observable:

$$\hat{L} = \arg \min_{1 \leq L \leq r_{\text{max}}} \|\mathbf{Z}_{\mathcal{V}} - |\mathbf{G}_L \mathbf{S}_{\mathcal{V}} + \mathbf{B}_{\mathcal{V}}|\|_F^2, \quad (11)$$

with $\mathbf{G}_L$ the estimate at order L from the fitting columns only, run for $T_{\text{sel}} = 20$ iterations. **No oracle rank is used anywhere in this paper**; the selector returns a single scalar $\hat{L}$ applied to all K columns. Order matters both ways: $\hat{L} < L$ truncates genuine components, a bias that does not vanish as noise decreases, while $\hat{L} > L$ admits noise and weakens the prior. Our selector systematically under-selects—true orders (7, 4, 6) typically give $\hat{L} = 5$, and in Section VII-C mean $\hat{L}$ reaches only 11.63 at $L = 16$.

Relation to the baseline. Deleting lines 5–7 of Alg. 3 yields baseline EM-GS exactly: both chain the same $\mathcal{T}_{\text{EM}}$ for the

Algorithm 3 HS-GS, and the held-out rank selection it uses

1: **function** HS-GS($\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; $\hat{L}, T$)
2: $\text{active} \leftarrow (\hat{L} < r_{\text{max}})$; $\mathbf{G} \leftarrow \emptyset$
3: **for** $t = 1..T$ **do**
4: $\mathbf{G} \leftarrow \mathcal{T}_{\text{EM}}(\mathbf{G})$ $\triangleright$ Alg. 1: *identical* to the baseline
5: **if** active **and** $t \bmod \Pi = 0$ **then**
6: $\mathbf{G}_{:,k} \leftarrow \mathcal{P}_{\hat{L}}(\mathbf{G}_{:,k})$, $k = 1..K$ $\triangleright$ Alg. 2
7: **end if**
8: **end for**
9: **return** $\mathbf{G}$
10: **end function**
11: **function** SELECTRANK($\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; ν, T_{sel})
12: $\mathcal{V} \leftarrow$ last $\lceil \nu P \rceil$ pilot columns; $\mathcal{F} \leftarrow$ the rest
13: **for** $L = 1..r_{\text{max}}$ **do**
14: $\mathbf{G}_L \leftarrow \text{HS-GS}(\mathbf{Z}_{:, \mathcal{F}}, \mathbf{S}_{:, \mathcal{F}}, \mathbf{B}_{:, \mathcal{F}}, \sigma^2; L, T_{\text{sel}})$
15: $J(L) \leftarrow \|\mathbf{Z}_{:, \mathcal{V}} - |\mathbf{G}_L \mathbf{S}_{:, \mathcal{V}} + \mathbf{B}_{:, \mathcal{V}}|\|_F^2$ $\triangleright$ columns never fitted
16: **end for**
17: **return** $\hat{L} = \arg \min_L J(L)$
18: **end function**

same T iterations from the same initialisation, with no early stopping in either. Setting $\hat{L} \geq r_{\text{max}}$ makes $\mathcal{P}_{\hat{L}}$ the identity (Alg. 2, line 2), so the two then agree *bit for bit*; this is checked numerically in §VI.

C. Complexity

One EM-GS update costs $O(NPK)$; the projection adds $K n_{\text{cz}}$ SVDs per iteration, $O(K n_{\text{cz}} N^3)$, and selection multiplies this by the r_{max} candidates. Against a *chained* EM-GS baseline (the correct reference, HS-GS issuing T single-iteration calls costing $1.005\text{--}1.038\times$ in overhead alone) HS-GS at fixed order costs $1.49, 1.32, 1.22\times$ at $N = 8, 16, 32$ —the ratio *falls* with N . Selection dominates the full estimator (57.7–85.5% of runtime), so shrinking the candidate set is the obvious practical target.

V. PERFORMANCE BENCHMARK**A. Likelihood and Fisher Information**

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$ the noiseless field at (n, p) is a known constant $\lambda = (\mathbf{GS} + \mathbf{B})_{n,p}$, so $z = |\lambda + w|$ is Rice distributed:

$$p(z | \lambda) = \frac{2z}{\sigma^2} \exp\left(-\frac{z^2 + |\lambda|^2}{\sigma^2}\right) I_0\left(\frac{2z|\lambda|}{\sigma^2}\right). \quad (12)$$

Crucially (12) depends on λ *only through* $a = |\lambda|$. Differentiating, with $\frac{d}{dx} \log I_0(x) = R(x)$,

$$\frac{\partial \log p}{\partial a} = \frac{2}{\sigma^2} [zR(\kappa) - a], \quad \kappa = \frac{2za}{\sigma^2}, \quad (13)$$

so one measurement carries scalar information $\beta(a, \sigma^2) = \mathbb{E}[(\partial_a \log p)^2]$, contributing to the Fisher matrix in real coordinates $\boldsymbol{\eta} = [\Re \text{vec } \mathbf{G}; \Im \text{vec } \mathbf{G}]$

$$\mathbf{J}_n = \sum_{p=1}^P 4\beta(a_{n,p}, \sigma^2) \mathbf{u}_{n,p} \mathbf{u}_{n,p}^T, \quad (14)$$

with $\mathbf{u}_{n,p} = \partial a_{n,p} / \partial \eta_n$. Each magnitude measurement therefore contributes rank-one information, not rank two. Crediting both quadratures overstates it and yields a bound too low by 0.17–4.64 dB across our 36 points. An identifiability check settles the form: with $K = 3$ and $P = 3$ the per-row problem is unidentifiable and (14) is correctly singular, whereas the rank-two construction claims full rank; Fig. 1 corroborates it, EM-GS sitting within 0.05 dB of the rank-one bound at high SNR. Because row n of $\mathbf{G}\mathbf{S}$ enters only measurement row n , $\mathbf{J}$ is block diagonal across receive elements—which is why the unconstrained bound, and any estimator attaining it, is flat in N under a normalised metric.

B. Constrained Bound on the Geometric Manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. HS-GS instead exploits (3), whose parameters $\phi = \{\psi_{\ell,k}, \Re\alpha_{\ell,k}, \Im\alpha_{\ell,k}\} \in \mathbb{R}^{3\sum_k L_k}$ number about 45, independent of N . With $\mathbf{D} = \partial\eta/\partial\phi$ the textbook form $\mathbf{D}(\mathbf{D}^T\mathbf{J}\mathbf{D})^{-1}\mathbf{D}^T$ is unusable, $\mathbf{D}$ being rank deficient: at $N = 8$ its mean numerical rank is 40.4 against ≈ 44.7 parameters, and 28.1% of realisations have $3\sum_k L_k > 2NK$, so $\phi \mapsto \mathbf{G}$ is not injective. We use the Gorman-Hero / Stoica-Ng tangent-space form [7], [8], $\mathbf{U}$ an orthonormal basis of $\text{range}(\mathbf{D})$ from SVD at tolerance 10^{-9} :

$$\text{CCRB} = \mathbf{U}(\mathbf{U}^T\mathbf{J}\mathbf{U})^{-1}\mathbf{U}^T, \quad (15)$$

which depends on the tangent *space* and is invariant to over-parameterisation. Worst conditioning of $\mathbf{U}^T\mathbf{J}\mathbf{U}$ is then 80.1 (at the degenerate $P = 6$), elsewhere below 35. Both bounds are averaged over the same 400 realisations as the estimator curves.

C. Bias Caveat

Both bounds constrain the covariance of *unbiased* estimators, whereas

$$\text{MSE} = \text{variance} + \text{bias}^2. \quad (16)$$

All three estimators run to a fixed budget with the shrinking update (4), and HS-GS additionally truncates rank; none is unbiased, so neither bound applies strictly. The unconstrained CRLB is therefore the reference for GS and EM-GS only—HS-GS uses prior information it does not model—and the CCRB, itself an unbiased-estimation bound, does not forbid a sufficiently biased structured estimator from falling below it. §VII-E reports both effects.

VI. SIMULATION SETUP

All results use the exact observation (1); the modulus is never linearised, which is verified per trial. NMSE is pooled as a ratio of sums,

$$\text{NMSE}_G = 10 \log_{10} \frac{\sum_t \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2}{\sum_t \|\mathbf{G}_t\|_F^2}, \quad (17)$$

never a mean of per-trial decibels: (17) estimates the ensemble quantity a CRLB bounds, whereas a mean-of-decibels is smaller by Jensen. The factor is 10, not 20, the arguments

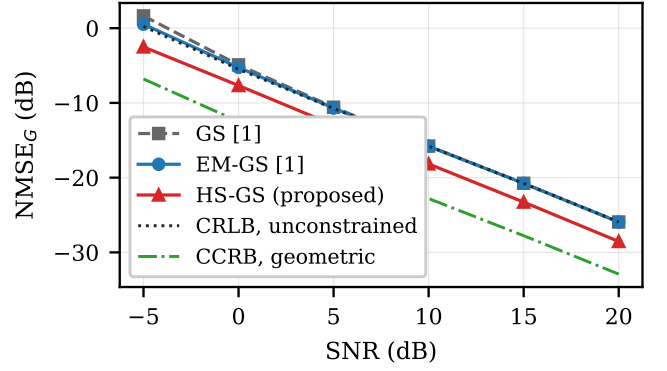


Fig. 1. NMSE versus SNR at $N = 32$, $P = 30$, $\text{RSR} = 12$ dB. EM-GS is indistinguishable from the unconstrained CRLB, so that bound is tight for the unstructured problem; HS-GS lies between the two, recovering part of the headroom identified by the geometric CCRB.

being energies. Within a point all estimators see bit-identical realisations, so the gain

$$\Delta_{\text{HS}} = 10 \log_{10} \frac{\sum_t \|\hat{\mathbf{G}}_t^{\text{EM}} - \mathbf{G}_t\|_F^2}{\sum_t \|\hat{\mathbf{G}}_t^{\text{HS}} - \mathbf{G}_t\|_F^2} \quad (18)$$

and its 95% paired bootstrap interval (2000 resamples) benefit from cancellation of the common channel draw; realisations are independent *across* points.

Two averaging domains, kept distinct. Within an operating point (17) sums energies first and converts last, averaging in the *linear* domain; across SNR (Fig. 5, bottom right) the mean is of the per-SNR decibel values, *after* conversion. Linear NMSE spans three orders of magnitude here, so a linear average across SNR is dominated by the lowest-SNR point ($\approx 70\%$ of the sum against 0.72% for the top three) and nearly blind to the high-SNR half—precisely where the projection hurts at $N = 8$.

Trial budget. Rank selection reruns the estimator once per candidate order, so a trial at $N = 32$ costs $\approx 8.5\times$ one at $N = 8$; the budget is tiered and every point reports its own count. §VII rests on 2.5×10^4 trials for the SNR and pilot-length sweeps, 8.4×10^3 on the $N \times \text{SNR}$ grid of Fig. 5, and 2.4×10^3 on the path-count sweep.

Verification. A standalone package reproduces the comparison and passes thirteen automated checks. Two are load-bearing for fairness and executed rather than asserted: each trial hands both estimators one *read-only* world ($\mathbf{G}, \mathbf{S}, \mathbf{B}, \mathbf{W}, \mathbf{Z}$), so they cannot diverge, and with the projection disabled HS-GS reproduces EM-GS to $\max|\text{diff}| = 0$ —the projection is provably the only difference. A third confirms §III: a noiseless L -path column has $\sigma_{L+1}/\sigma_1 \approx 7 \times 10^{-16}$.

VII. RESULTS AND DISCUSSION

A. Main Result and Array-Size Scaling

First the operating point. EM-GS improves on biased GS by 1.31 dB at $\text{RSR} = 0$ dB, collapsing to 0.113 dB at 12 dB and 0.002 dB at 24 dB as $R(\kappa) \rightarrow 1$ and the two provably coincide. $\text{RSR} = 12$ dB, from [1], keeps the baselines separable, so the gains below are attributable to the projection, not the choice of exact step. Fig. 1 shows the comparison at $N = 32$,

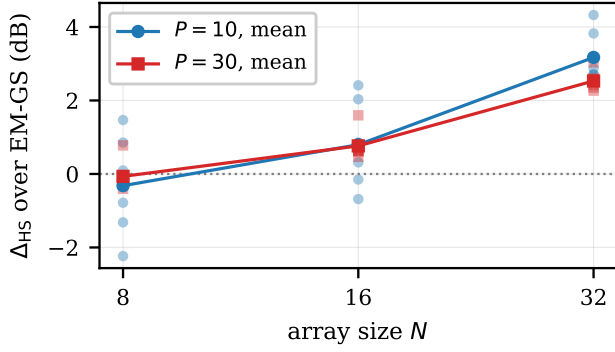


Fig. 2. Paired HS-GS gain over EM-GS versus array size. Faint markers are the six individual SNR operating points, solid lines their mean. The gain grows monotonically with N ; at $N = 8$ the points straddle zero, so no single scalar summarises that array size.

$P = 30$. EM-GS tracks the unconstrained CRLB to within 0.05 dB for $\text{SNR} \geq 5$ dB, so it is close to efficient for the unstructured problem and the corrected bound is credible. The geometric CCRB lies 0.87–9.98 dB below it, quantifying what the structure is worth in principle; HS-GS occupies part of that gap, improving on EM-GS by 2.27–3.04 dB across the sweep.

Fig. 2 isolates the scaling. Averaged over the twelve operating points at each array size the gain is -0.19 , $+0.78$, $+2.85$ dB for $N = 8, 16, 32$, win rate 33.5, 74.0, 95.3%, constraint active in 57.5, 92.7, 99.6% of trials; the largest single point is 4.33 dB. Under the same averaging EM-GS varies by only 0.012 dB, as §V requires: the N -dependence belongs to the structural prior, not a degrading baseline.

The mechanism is the ratio of path count to rank ceiling (9): at $N = 8$, $r_{\max} = 4$ while $L_k \in \{3, \dots, 7\}$, so the projection is inactive in 43% of trials and where active often truncates real components, whereas at $N = 32$, $r_{\max} = 16 \gg L_k$ and it is active in 99.6%. At $N = 8$ the sign of any scalar summary depends on the pooling—the per-point mean is -0.19 dB, energy-pooling gives $+0.81$ dB—because HS-GS wins at low SNR and loses at high SNR there. That high-SNR loss is a bias-dominated floor: truncation bias does not shrink with noise, so as variance falls the trade stops paying. At $N = 16$ and 32 every pooling agrees in sign.

B. Controlled Path Count: When and Why the Method Works

Every experiment above draws L_k at random, so none isolates path count. Fig. 3 fixes $L_k = L$ across users and sweeps it to $r_{\max}$; the gain decays strictly monotonically,

7.04, 3.56, 1.79, 1.04, 0.58, 0.27, 0.05, -0.12 dB

for $L = 2, 4, \dots, 16$, with the per-trial win rate falling from 100% to 45.0% and EM-GS flat to 0.19 dB across the sweep. The interpretation is direct: when $L \ll r_{\max}$ the lifting carries substantial redundant structure, so projecting suppresses error that does not share it; as $L \rightarrow r_{\max}$ the prior admits almost every vector and the benefit disappears.

Two details matter. The null is reached at $L = 14$ (gain $+0.05$ dB, CI $[-0.05, +0.14]$), one grid point *before* the ceiling, and at $L = 16$ it is small but significantly negative (-0.12 dB, CI $[-0.21, -0.04]$). The selector explains this,

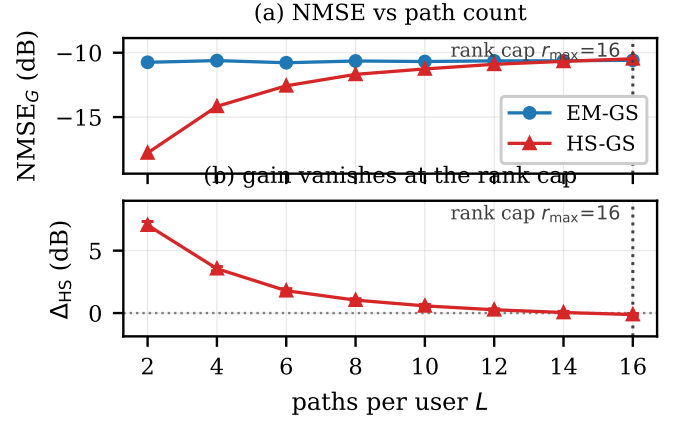


Fig. 3. Controlled path count ($N = 32$, $P = 30$, $\text{SNR} = 5$ dB, $\text{RSR} = 12$ dB, 300 trials per point, L fixed and identical across users). (a) EM-GS is flat in L —path count does not affect an estimator that ignores structure—which confirms the sweep is not confounded; HS-GS rises to meet it. (b) The paired gain decays monotonically to zero as L approaches the rank ceiling $r_{\max} = \lceil N/2 \rceil = 16$; the 95% paired bootstrap band is plotted but is narrower than the markers. The hypothesis producing this figure was recorded before the experiment was run.

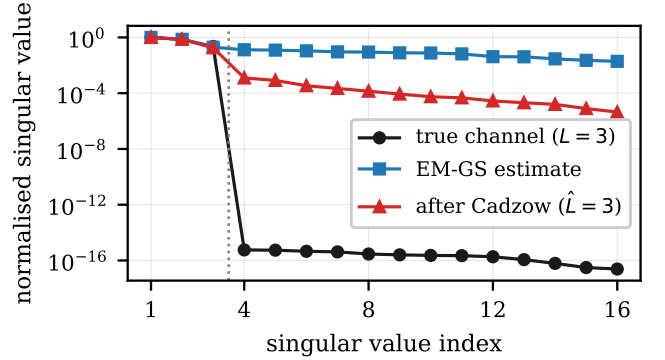


Fig. 4. Normalised Hankel singular values for one channel column, $N = 32$, $L = 3$, $\text{SNR} = 5$ dB. The true channel is exactly rank L , the EM-GS estimate is full rank, and Cadzow restores the cliff only approximately, to $\sim 10^{-3}$.

not the theory: mean $\hat{L}$ reaches only 11.63 at $L = 16$, so the constraint is still active in 89.3% of trials and there discards genuine components—the $\hat{L} < L$ regime of §IV-B. Second, this rules out the obvious rival: generic denoising would persist at large L , where the estimate is equally noisy, not vanish exactly where the prior becomes vacuous.

C. Structural Diagnostic

Fig. 4 shows why the prior exists and what the projection achieves. The true channel’s Hankel spectrum falls to $\sim 10^{-16}$ immediately after index L : the low-rank property is exact, as (8) states. The EM-GS estimate has no such cliff—noise fills the spectrum, so the estimate is not a sum of L exponentials. After Cadzow a cliff reappears at the correct index but only to $\sim 10^{-3}$; that shortfall is the four-sweep projection being approximate, and it upper-bounds how much structure can be imposed—consistent with HS-GS attaining only part of the CCRB headroom.

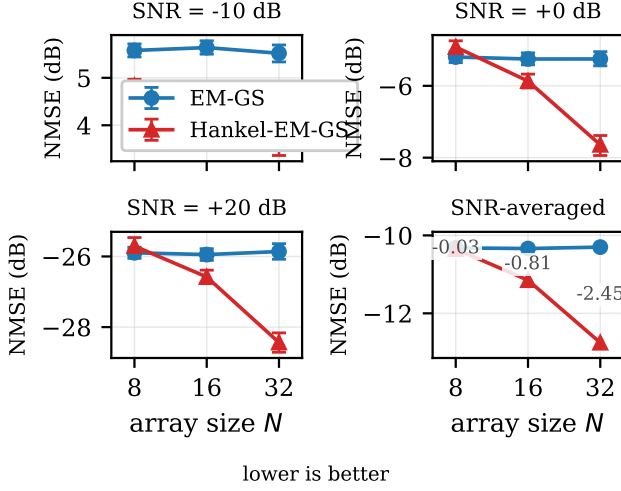


Fig. 5. Absolute channel NMSE versus array size: three SNR regimes and, bottom right, the SNR-averaged summary (mean of the seven per-SNR values taken *after* conversion to decibels, §VI; inset labels give HS-GS minus EM-GS). $K = 3$, $P = 30$, RSR = 12 dB, $L_k \sim \mathcal{U}\{3..7\}$; 600/400/200 paired trials per point at $N = 8/16/32$; error bars 95% bootstrap.

D. Absolute NMSE of Both Estimators versus N

Fig. 2 plots the *difference* Δ_{HS} , leaving open how each estimator behaves alone; Fig. 5 plots the two NMSE curves themselves, at $P = 30$ over an SNR grid extended to -10 dB.

Two facts separate cleanly. **EM-GS is flat in N :** SNR-averaged -10.321 , -10.337 , -10.301 dB at $N = 8, 16, 32$, a spread of 0.036 dB, and equally flat inside every panel—a larger array buys the unstructured estimator essentially nothing, as the block-diagonal Fisher structure requires. **HS-GS is the only curve that moves:** -10.350 , -11.149 , -12.753 dB, so the gap widens monotonically, $0.03 \rightarrow 0.81 \rightarrow 2.45$ dB, at every SNR tested.

The $N = 8$ panel needs its own sentence: the SNR-averaged 0.03 dB conceals a sign change, HS-GS gaining 0.744 dB at -10 dB but losing 0.190 – 0.403 dB at 0 dB and above, intervals excluding zero both ways. $N = 8$ is a mixed result, not a small positive one.

E. Distance to the Bounds, and Crossings

HS-GS lies above the geometric CCRB at 34 of 36 points, by 0.32 – 8.85 dB, and the gap *grows* with N ($+0.32$ – 1.41 dB at $N = 8$, $+4.29$ – 4.82 dB at $N = 32$, $P = 30$): it captures more structural gain in absolute terms as the array grows but a smaller share of what is available.

Curves do fall below the plotted bounds, in two distinct senses. HS-GS lies below the *unconstrained* CRLB at 26 of 36 points, by up to 6.94 dB, but this is no violation: that bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters, whereas HS-GS exploits a prior it does not model. Its proper bound is the CCRB, respected at 34 of 36. The remaining crossings are consistent with bias, as (16) permits, and track the measured shrinkage $c = \langle \hat{\mathbf{G}}, \mathbf{G} \rangle / \|\mathbf{G}\|_F^2$ in both magnitude and location: GS and EM-GS cross by up to 2.62 dB at -5 dB where $c_{\text{EM}} = 0.944$, but by at most 0.05 dB for $\text{SNR} \geq 15$ dB

where $c \rightarrow 1$; HS-GS falls below the CCRB at exactly the two points of strongest shrinkage ($c_{\text{HS}} = 0.561, 0.776$). We do not claim bias accounts for every crossing.

VIII. LIMITATIONS AND FUTURE WORK

All results here come from the i.i.d. uniform-AoA, equal-per-path-power, flat geometric model of §II, so the defensible claim is model-conditional: *under the tested sparse geometric multipath model, exploiting the low-rank Hankel structure of the ULA response improves channel-estimation NMSE relative to EM-GS, by a mean of 2.85 dB at $N = 32$* . Nothing here establishes robustness under clustered propagation, and the decisive next experiment is to repeat the principal comparison under a clustered 3GPP TR 38.901 generator [9]—a falsification test, not a confirmation exercise. Under clustering the z_ℓ of a cluster are tightly packed, so $\mathcal{H}(\mathbf{g})$ is only *numerically* low rank, with a decaying spectrum rather than an exact cliff, and hard truncation becomes a bias-inducing approximation. Since the $N = 8$ high-SNR degradation already traces to truncation bias, clustering could plausibly weaken or eliminate the gain.

Further, a single scalar $\hat{L}$ is imposed on all K users despite their differing orders; (10) has no convergence guarantee, its constraint set being non-convex and its projection approximate; and all users have equal power, so near-far effects are untested.

IX. CONCLUSION

A sparse geometric ULA channel makes each column’s Hankel lifting rank-deficient, and this can be enforced inside an exact biased phase-retrieval iteration without linearising the magnitude observation or estimating any angle. HS-GS improves on EM-GS by a mean of 2.85 dB at $N = 32$ under the tested model; absolute NMSE makes the asymmetry explicit, the unstructured baselines being flat in array size so HS-GS alone converts a larger aperture into lower normalised error. That the gap widens with N is an *empirical* finding over $N \in \{8, 16, 32\}$, consistent with $\lceil N/2 \rceil$ growing past L_k but not proved beyond the sizes simulated; the benefit is conditional, negative at $N = 8$ above 0 dB SNR. The path-count sweep delimits the mechanism: the gain scales with r_{max}/L and vanishes at the ceiling. Validation under clustered propagation remains the essential next step.

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